legal symbol name (the lead byte is in the range *0xC0–0xFD*, and continuation bytes are in the range *0x80–0xBF*), and any valid single-byte UTF-8 character simply needs to be in the range a–z, or A–Z, or 0–9, or an underscore _.

It's worth noting that although PHP's symbol naming rules allow for UTF-8 encoded text, UTF-8 is not enforced. It is possible to use a single byte in the *0x80–0xFF* range as a symbol, which we can demonstrate with a hex editor (which allows binary-level editing). Take the following text and save it to a file called `foo.php` (encoding doesn't matter; we're about to break it):

```
<?php

$x = 1;
var_dump($x);
```

Now, using a hex editor, edit the file. The example below uses a MacOS hex editor called Hex Fiend, which is available for free. Find both occurrences of the sequence `2478`…

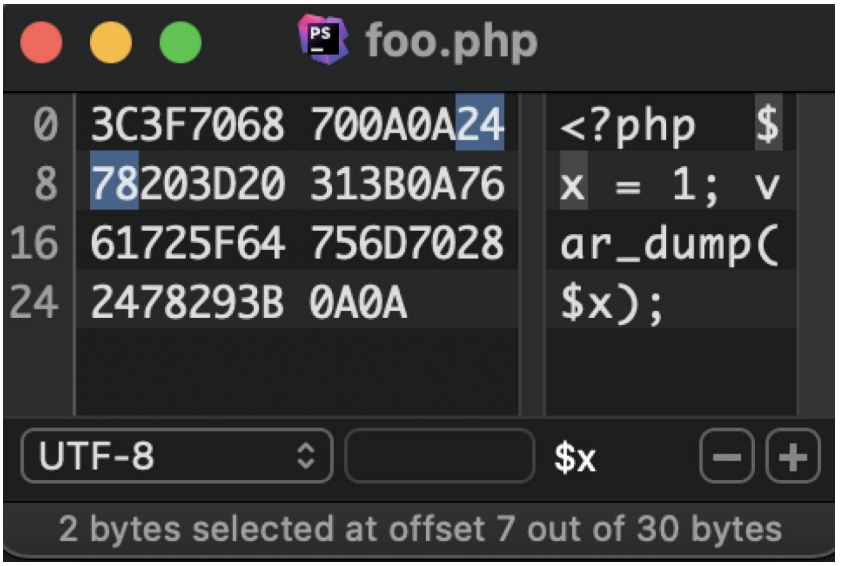


Figure 8-1. *hex edit of raw byte values before…*

… and change them to `2480` …